

Conditional Probability and Independence

Section 1: Recap of Probability

If you roll a fair die, the chance of getting a 4 is **1 out of 6** — simple.

But real-world data is not that clean. In ML, we often deal with situations like:

“What’s the chance a person buys pizza if it’s Friday night?”

That’s no longer simple probability — it’s *conditional*.

Section 2: What is Conditional Probability?

Conditional probability measures how likely one event is *given* that another event has occurred.

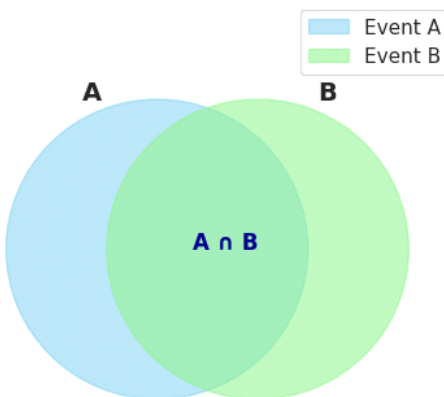
Mathematically:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Read it as:

“The probability of A given B equals the probability of both A and B happening, divided by the probability of B.”

Conditional Probability: Overlapping Events



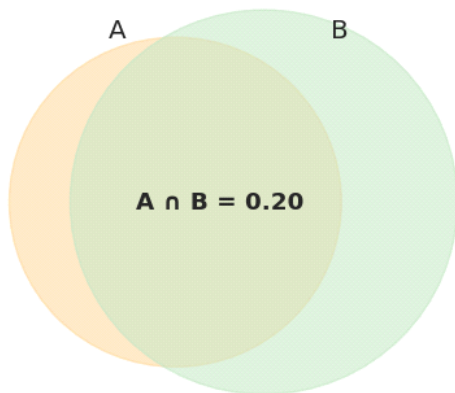
Example:

Suppose in a dataset of 1,000 people:

- 300 like pizza → $P(A)=0.3$
- 500 are students → $P(B)=0.5$
- 200 are students who like pizza → $P(A \cap B)=0.2$

Pizza & Students — Two-circle Venn (schematic)

Pizza (A)
 Students (B)



Then:

$$P(A|B) = \frac{0.2}{0.5} = 0.4$$

That means *if you know someone is a student, the probability they like pizza jumps from 0.3 to 0.4.*

Now, let's visualize this.

Imagine a class of 100 students.

- 60 of them like coffee.
- 30 of them like both coffee and tea.

Coffee vs Tea — 2×2 Contingency with Conditional Highlight

	Tea	Not Tea	Total
Coffee	30	30	60
Not Coffee	10	30	40

$$P(\text{Coffee} \mid \text{Tea}) = 30 / 40 = 0.75$$

Now, if I ask — “What’s the probability that a student likes coffee **given that** they already like tea?”

We’re focusing only on the group of tea lovers — maybe that’s 40 students.
Out of those 40, 30 like coffee too.

So:

$$P(\text{Coffee}|\text{Tea})=30/40=0.75$$

That’s **75%**.

Section 3: Independence

If knowing one event doesn't change the probability of the other, they're **independent**.

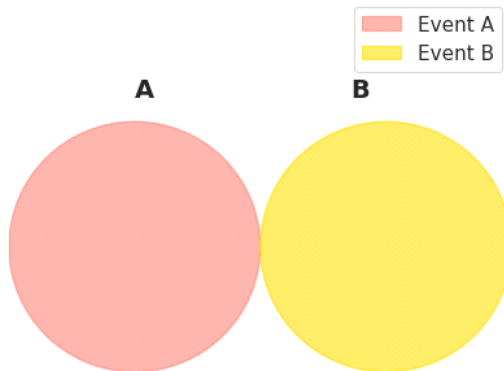
Mathematically:

$$P(A|B) = P(A)$$

and equivalently,

$$P(A \cap B) = P(A) \times P(B)$$

Independent Events: No Overlap



Example:

If gender and pizza preference are unrelated in your data, then whether someone's male or female doesn't change the probability of liking pizza.

So, if $P(\text{LikesPizza}) = 0.6$ and $P(\text{WearsGlasses}) = 0.3$,

then if these are independent,

$$P(\text{Both}) = 0.6 \times 0.3 = 0.18$$

— meaning 18% of people are expected to both like pizza and wear glasses, *just by chance*.

In ML, **independence** helps simplify complex problems.

For example, **Naive Bayes classifier** assumes that all features are independent given the class — even though they aren't truly independent — but it still works surprisingly well.

Section 4: Conditional Independence

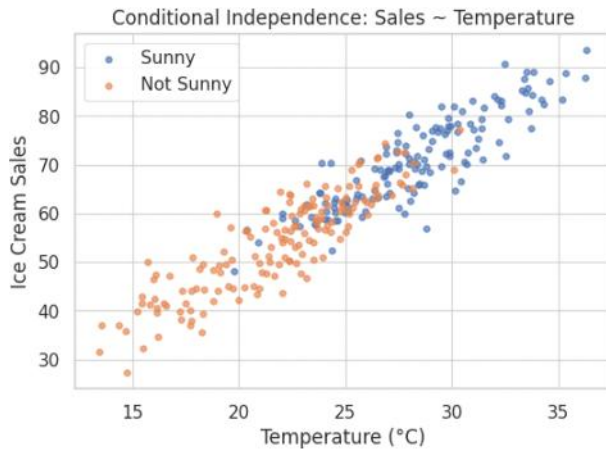
Two events A and B are *conditionally independent* given C if:

$$P(A|B,C) = P(A|C)$$

That means once you know C, B gives no extra information about A.

Example:

Suppose you're predicting whether someone will buy ice cream (A) based on temperature (C) and whether it's sunny (B). If you already know the temperature, the "sunny" info adds almost nothing — so A and B are conditionally independent given C.



In many ML models (like **Naive Bayes**, **Bayesian Networks**, or **Hidden Markov Models**), we assume certain variables are conditionally independent to simplify calculations.

If A and B were *always* independent, we could say

$$P(A,B)=P(A)P(B)$$

But real-world data often has dependencies. So instead, we assume independence **given** something else (a “context” variable like C).

Let’s say you have:

- **A:** Whether a student passes the exam
- **B:** Whether they studied with friends
- **C:** Total hours studied

If you already know the total study hours (C), then whether they studied alone or with friends (B) adds very little extra info about passing (A). So empirically you’d find:

$$P(\text{Pass} \mid \text{StudiedWithFriends}, \text{Hours}) \approx P(\text{Pass} \mid \text{Hours})$$

That’s **conditional independence** in action.

Section 5: Why It Matters in ML

Conditional probability allows models to make predictions under uncertainty.

Examples:

- **Spam detection:** $P(\text{Spam} \mid \text{Contains “lottery”})$
- **Medical diagnosis:** $P(\text{Disease} \mid \text{Test Positive})$
- **Recommender systems:** $P(\text{User likes item} \mid \text{User liked similar items})$

When combined with independence assumptions, we get fast, interpretable models like **Naive Bayes**, still used in text classification today.

Section 6: Wrap-Up

So, to summarize:

- Conditional probability tells us *how one event affects another*.
- Independence means *no effect*.
- Conditional independence helps simplify large models.
These ideas form the foundation of many ML algorithms that learn from *patterns and uncertainty*.

Next time your model makes a “smart guess,” remember — it’s all probability at play.